

SUDHANVA DEVANATHAN

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Work Experience

BlackRock

Jan 2021 – July 2024

Software Engineer II

Mumbai, India

- Delivered analytics across **100K+ portfolios** to **130+ clients** for a new product offering by building a **Python** orchestration pipeline from scratch on the Aladdin platform that processes **100M+ rows** nightly.
- Cut database connections by **80%** across 130+ client environments (~500K gRPC calls/day) by designing a **Java/Spring Boot gRPC** config service that serves configs from memory, kept consistent on restart via exactly-once **Kafka**.
- Raised on-time failure detection by **40%** by building an event-driven monitoring system in **Python (asyncio)** whose distributed pollers track financial models concurrently and publish metrics over gRPC into Splunk.
- Sped up the report orchestrator **5x**, with **35%** less memory, **30%** fewer DB queries and **30%** lower client execution time, by owning its rewrite from **Perl** into a config-driven async **Python** service, de-risking cutover with a parallel run.
- Onboarded **30+ data sources** onto a shared **Go** orchestration engine used by multiple teams by shipping 5+ DSL and ingestion features and its **Helm** charts for OpenShift/Kubernetes and Azure, supporting the C++-to-Go rewrite.
- Raised code coverage from **80% to 95%** in 30% of team repos by leading the CI/CD migration to **Azure DevOps**.
- Held client data-delivery SLAs on **L2/3 on-call** by building fallback mechanisms and diagnosing production failures.

BlackRock

July 2018 – Dec 2020

Software Engineer I

Mumbai, India

- Reduced average job time-to-start by **30%** by replacing fixed-schedule cron with an **event-driven** execution layer that triggers each risk and index calculation once its dependency tree resolves.
- Cut analytics job latency by **20%** by designing a **memory-aware scheduler** that routes jobs by memory history.
- Achieved a **98% on-call success rate** by resolving risk-analytics incidents as L1/2 on-call.

Personal Projects

Distributed Key-Value Store | *Go, Raft, gRPC, Protobuf*

- Kept a **5-node** key-value cluster strongly consistent and fault-tolerant by implementing **Raft** from scratch in Go (leader election, log replication), sharded by **consistent hashing** and verified race-free under Go's **race detector**.
- Cut memory **~60%** in long-running sessions by adding multi-level compaction to its from-scratch storage layer (CRC32 **write-ahead log**, **LSM tree** with SSTables, sparse index and Bloom filters, crash recovery via WAL replay).

rusty-redis: Redis-Compatible In-Memory Server | *Rust, Tokio, RESP*

- Served Redis clients with **byte-for-byte Redis 8.10.1** replies (~100 differential tests in CI) by architecting a **Rust/Tokio** server where one task owns every key, making commands atomic **without locks**.
- Extended it to **29 commands** with WATCH transactions, key expiry and disconnect-safe blocking reads.
- Measured **66–80%** of Redis 8.10.1 throughput unpipelined and **39–97%** pipelined (M1 Pro) by building a harness that benchmarks both servers in fresh processes; unpipelined throughput held flat at ~104K ops/s, pointing to round-trip cost.

probabilistic-chord: Chord DHT with Probabilistic Filter Routing | *C++11*

- Cut maintenance-message overhead **~50%** during stable periods while holding **$O(\log n)$** lookup hops in a **C++11** Chord DHT by designing three pluggable routing strategies (finger table, **Cuckoo** and **Quotient** filters).
- Benchmarked hops, latency and false-positive sensitivity from 8 to **1024 nodes** running as separate OS processes.

Education

Northeastern University

Boston, MA

MS in Computer Science, GPA 3.84; Graduate TA, CS 5010 (Program Design Paradigm)

Sept 2024 – May 2026

Vellore Institute of Technology | *BTech in Computer Science and Engineering*

2018

Technical Skills

Languages: Go, Java, Python, Rust, C++, SQL, Shell

Distributed Systems: Raft, Consistent Hashing, LSM Trees, WAL, Exactly-Once Delivery, Event-Driven Architecture

Storage & Databases: PostgreSQL, Cassandra (schema design, ingestion), Redis, MySQL

Cloud & DevOps: Azure, Kubernetes (OpenShift), Helm, Docker, Azure DevOps, GitHub Actions, Linux/Unix

Frameworks & Tools: Spring Boot, gRPC, Protobuf, Kafka, Spark, Tokio, asyncio, pytest, JUnit

AI-Assisted Development: Claude Code, Codex, Cursor; agent output validated with fuzz tests (personal projects)